

homework₁

June 12, 2026

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(1) If R is a ring, show that $M_n(R)$ is a ring under the usual operations of matrix addition and multiplication.

(2) Let $(G, +)$ be an Abelian group. An endomorphism of G is a group homomorphism $\varphi : G \rightarrow G$. Let $\text{End}(G)$ denote the set of all endomorphisms on G . Given $\varphi, \psi \in \text{End}(G)$, define

2. Let R be a ring, S be a subring of R and let I be an ideal of R .
- (a) Prove that $S + I := \{s + a : s \in S, a \in I\}$ is a subring of R .
 - (b) Prove that $I \triangleleft S + I$.
 - (c) Prove that $S \cap I \triangleleft S$.
 - (d) Prove that

$$\frac{S + I}{I} \cong \frac{S}{S \cap I}$$

[Hint: Define $\varphi : S \rightarrow (S + I)/I$ by $s \mapsto s + I$.]

3. Let $D \in \mathbb{Q}$ be a rational number that is not a perfect square. Define

$$= (A \cdot B) \cdot C$$

Distributive Property:

$$\begin{aligned} AB + AC &= \left(\sum_{k=1}^n a_{ik}b_{kj} + \sum_{k=1}^n a_{ik}c_{kj} \right)_{n \times n} \\ &= \left(\sum_{k=1}^n a_{ik}(b_{kj} + c_{kj}) \right)_{n \times n} \\ &= A(B + C) \end{aligned}$$

Similarly, we can show $(A + B)C = AC + BC$.

□

$$(\varphi \times \psi)(g + h) = \varphi(\psi(g + h)) = \varphi(\psi(g) + \psi(h))$$

$$= \varphi(\psi(g)) + \varphi(\psi(h)) = (\varphi \times \psi)(g) + (\varphi \times \psi)(h)$$

Thus, $\varphi \times \psi \in \text{End}(G)$. □

(b)

Theorem 0.1. Prove $(\text{End}(G), +, \times)$ is a ring.

Proof. As G is an Abelian group, $(\text{End}(G$

Proof. $0 \cdot a = (a - a) \cdot a = a^2 - a^2 = 0,$

$$a \cdot 0 = a \cdot (a - a) = a^2 - a^2 = 0. \quad \square$$

b) Show: $(na)b = n(ab) = a(nb).$

Proof. Base case ($n=1$): Trivial.

Induction Hypothesis: $(na)b = n(ab) = a(nb).$

Proof. Assume S and T to be non-empty. □

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Let $a, b \in ST$ such that $a = \sum_{i=1}^{n_1} s_i t_i$ and $b = \sum_{i=1}^{n_2} s'_i t'_i$.

Without loss of generality, assume $n_2 > n_1$. Then,

$$b - a = \sum_{i=1}^{n_2} s'_i t'_i - \sum_{i=1$$

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Given $m, n \in \mathbb{Z}$ and $d = \gcd(m, n)$.

Definition 0.2. Let I be the ideal generated by $\{m, n\}$.

Theorem 0.2. Show that $I = (d)$.

Proof. Consider $(X) = \bigcap_{J \in \mathcal{F}} J$, where

$$\mathcal{F} = \{J \triangleleft \$$

Consider $x_0 = \frac{a'+a}{2}$. Then,

$$g(x_0) = 0 \quad \text{and} \quad f(x_0) \neq 0 \implies f \times g(x_0) \neq 0.$$

This leads to a contradiction since $f \times g = f$ implies $f \times g(x_0) = f(x_0) \neq 0$ but $g(x_0) = 0$ implies $f \times g(x_0) = 0$. Hence, no unit exists in $C_c(\mathbb{R})$. $\square$

Given the handwritten content, —

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Hence, as $s \in S \cap I \implies \varphi(s) = 0 + I \implies S \cap I \subseteq \ker \varphi$.

If $s \in \ker \varphi, \varphi(s) = I \implies s \in S$ and $s \in I \implies s \in S \cap I$.

$$\therefore S \cap I = \ker \varphi.$$

Proof. (cd) Prove: $\frac{S+I}{I} \cong \frac{S}{S \cap I}$

(1) Well definedness

$$s_1 = s_2 \implies s_1 + I = s_2 + I$$

(2) φ is a ring homomorphism

$$\varphi(s + s') = s + s' + I = (s + I) + (s' + I) = \varphi(s) + \varphi(s')$$

$$\varphi(ss') = ss' + I = (s + I)(s' + I) = \varphi(s)\varphi(s')$$

(3) φ is surjective (By construction)

(4) $\ker \varphi = S \cap I$

$$\text{If } s \in S \cap I, \text{ now } \varphi(s) = 0 + I \text{ as } s \in S \cap I$$

Hence, $S \cap I \subseteq \ker \varphi$.

If $s \in \ker \varphi, \varphi(s) = I$, now $s \in S$ and $s \in I \implies s \in S \cap I$.

$$\therefore S \cap I = \ker \varphi.$$

By First Isomorphism Theorem,

$$\frac{S+I}{I} \cong \frac{S}{S \cap I}.$$

□

Given the content in the image, —

1. Given: $D \in \mathbb{Q}$ be a rational number and is NOT a perfect square.

$$R := \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

(a) *Proof.* Show: R is a subring of $\mathbb{C}$.

Observe, $R \neq \emptyset$ as $0 \in R$.

Consider $x, y \in R$, where $x = a_1 + b_1\sqrt{D}$ and $y = a_2 + b_2\sqrt{D}$; $a_1, b_1, a_2, b_2 \in \mathbb{Q}$.

$$x - y = (a_1 - a_2) + (b_1 - b_2)\sqrt{D} \in R \quad \text{as} \quad a_1 - a_2 \in \mathbb{Q} \quad \text{and} \quad b_1 - b_2 \in \mathbb{Q}.$$

Thus, $(R, +)$ is a subgroup of $(\mathbb{C}, +)$.

Now, look at

$$xy = (a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D}) = a_1a_2 + b_1b_2D + (a_2b_1 + a_1b_2)\sqrt{D}.$$

Since $a_1a_2 + b_1b_2D \in \mathbb{Q}$ and $a_2b_1 + a_1b_2 \in \mathbb{Q}$, it follows that $xy \in R$.

Hence, $(R, +, \cdot)$ is a subring of $(\mathbb{C}, +, \cdot)$. □

(b)

Theorem 0.6. If $(a + b\sqrt{D}) \neq 0$, then show that $a^2 - b^2D \neq 0$.

Proof. If $(a + b\sqrt{D}) \neq 0$, then $a \neq 0$ or $b \neq 0$.

Hence, $a - b\sqrt{D} \neq 0$. Now,

$$(a + b\sqrt{D})(a - b\sqrt{D}) \neq 0 \implies a^2 - b^2D \neq 0.$$

□

(c)

Theorem 0.7. Prove: R is a field.

Proof. Observe R is a commutative, unital ring.

Moreover, R is a division ring as for $a + b\sqrt{D} \neq 0$, then

$$\frac{a - b\sqrt{D}}{a^2 - b^2D}$$

is the inverse with respect to multiplication.

Now, R is a commutative division ring, which is a field. □

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Given R is a unital ring,

$$Z(CR) := \{a \in R \mid ab = ba \text{ for all } b \in R\}.$$

Proof. Observe, $Z(CR) \neq \emptyset$ as $1 \in Z(CR)$.

Now, consider $a, a' \in Z(CR)$.

For any $b \in R$, $(a - a')b = ab - a'b = ba - ba' = b(a - a') \implies (Z(CR), +)$ is a subgroup of R .

For any $b \in R$, observe $aa'b = aba' = baa' \implies aa' \in Z(CR)$.

Hence, $Z(CR)$ is a subring.

□